Internal Transfers, Hiring, and Career Trajectories *

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Abstract

How do large organizations replace workers lost to military conscription? We study this question using a novel personnel database covering all employees and identified draftees in the wartime Tokyo civil service. Under a conditional exogeneity design with pre-war balance checks, each draft-induced exit causally increases internal transfers by about 0.41 and external hires by about 0.22, with reallocation concentrated across sections within the same department. Offices also promote remaining workers (8.7 pp) and offer bonus pay to retain scarce staff. Descriptive evidence suggests the central HR team sources transferees from large offices with pre-existing organizational slack. Transferred workers outperform peers at both origin and destination offices on retention, advancement, and tenure—observable attributes do not predict selection into transfer, though unobserved selection cannot be ruled out. Both transfers and new hires are predominantly male, leaving office gender composition unchanged; women's entry into the civil service instead coincides with voluntary male exits amid a heated wartime labor market.

Keywords: Military conscription, Internal labor markets, Worker reallocation, Personnel economics, Wartime Japan.

JEL Codes: M51, M54, N45, J45.

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1 Introduction

We study the impact of military-draft-induced displacement on adjacent workers and its long-run effects. Conflicts and wars displace a large share of the workforce from incumbent positions through military drafting (Acemoglu et al. (2004)) and allocation across sectors (Garin (2025)). Such displacement induces workplaces to find replacements from the existing workforce and to hire from external labor markets (Acemoglu et al. (2004), Goldin (2006)). However, searching for workers during wartime is difficult because conscription targets young eligible workers who are more likely to have already been conscripted into the military, while wartime government spending overheats hiring efforts and raises hiring costs. Such temporary assignments may have permanent effects on the careers of appointees as well as on their future management of workers under their supervision.

We study how vacancies due to military drafting caused the reallocation of staff in organizationally adjacent offices in the Tokyo civil service during World War II. The Japanese military mobilized the majority of eligible men to the battlefields, and a 1945 bill sent 65% of the working population into the war, with conscription concentrated among younger workers. We compile a full 35-year personnel record of the civil service with positional and occupational assignments for each worker, as well as their drafting status during the war period. The civil service consisted of approximately 40,000 employees per year, distributed across 2,000 unique occupation-position units spread throughout Tokyo prefecture. First, we identify staff reallocation due to drafting by comparing organizationally and geographically adjacent office units while controlling for differences in their occupational-role composition. We study how the workplaces of draftees used promotions and bonus salaries to retain existing workers and transferred workers from adjacent offices, both geographically and organizationally, to fill vacancies. The national government of Japan kept its military drafting process hidden from eligible male workers to prevent them from sorting into positions exempt from conscription. We observe that drafted workers were therefore assigned to a wide range of ranks and office units upon conscription[1]. We distinguish between distances across

[1] 1,422 of 1,924 position–office cells across four years.

offices and differences in task content across offices.

Second, we document the transferred workers' subsequent careers in the civil service, promotional outcomes for peers at rank levels similar to those of draft-induced transferees, and transferees' managerial influence over their subordinates. The Tokyo civil service had a strict rotational policy that restricted the maximum length of time an employee could stay in a fixed position. We test the outcomes of short-term assignments at the destination office and compare draft-induced transferees' average salaries with those of their immediate peers in the same promotional candidate group.

We find that retention of existing workers and transfers from adjacent office units were the dominant sources of labor for replacing draft-induced vacancies. Each additional draft-induced exit generated about twice as many internal transfers (0.41) as external hires (0.22). The response is driven mainly by transfers from a different section within the same department (about 0.29 workers) and from different departments (about 0.14 workers). Alongside recruiting workers from internal and external sources, offices mitigated the disruption caused by drafting by promoting workers in lower-ranked positions and offering bonus salaries to the remaining staff to retain them. A one-unit increase in the number of drafted workers increased the probability that a non-drafted coworker was promoted by 8.7 percentage points. An additional draft also increased the number of existing staff receiving bonus salary (*rinji tokubetsu*) allowances by 0.14 units. We find that the retention probability for existing workers was slightly higher in draft-impacted offices (0.9 percentage points per draft), but the estimate is statistically insignificant.

Within donor offices, transfers were not predicted by workers' gender, experience, productivity, or rank. The mojority of the workers that was transferred to a draft impacted office received an improvement in salaries or rank relative to 2 years before the transfer. This accrued from the Tokyo civil service absorbing the munincipal governments in Tokyo and offering new contracts and laying off nearly 40% of the workforce in 1943. We find that office units selected for transferring a worker were those with organizational slack. Office units associated with transferring out a worker were more likely to have higher turn over of staff, due to their large size and we do not observe a increase in numbers of new hires after losing a worker. The hiring gap between donor and non-donor offices predates the transfer itself, with 5.1 additional hires in the prior year, consistent with pre-existing organizational capacity rather than vacancy-chain hiring in response to the outbound transfer. Within a office, we find offices located in distant jurisdictions, both organizationally and geographically, relied more on external signals such as productivity when choosing whom to select, while short-distance transfers relied on tenure and experience. Productive workers were more likely to get selected to transfer to a drafted office by [] percent in remote offices, but only [] percent in adjacent offices. Tenure strongly predicts transfer into a drafted office in adjant offices by [] percent, while only [] percent in remote offices.

Transferred workers stayed employed longer in the civil service after the war than peers at both destination and origin offices and experienced improvements in their occupational rank. Compared with destination and origin peers, transferred workers had 10.5 and 16.3 percentage point higher postwar survival rates, 0.12 and 0.18 units greater rank advancement, and 1.7 and 3.4 additional years of tenure, respectively. These differences remain after comparing them against origin peers who were transferred to an office unit with non-drafted vacancies and controlling for worker productivity. Retention length varied by the distance between origin and destination office units, as well as by similarities in their composition of occupational roles. Draft-induced transferees were later associated with managing offices that promoted subordinates with higher educational attainment. Additionally, their offices were more likely to select men than women for promotion into more senior roles. The higher retention rate of draft-induced workers reduced promotion opportunities for lower-ranked subordinates, as lower-ranked workers in other offices experienced much faster promotion tracks.

The results underscore how workplace disruptions due to military drafting were met by workplaces seeking replacements from internal sources rather than from the external market. The expertise of workers constitutes a valuable asset for workplaces, and replacing workers with external hires imposes an additional cost (Oi (1962), Jäger et al. (2025)). The composition of the neighboring organizational structure predicts the success of finding a replacement to fill the role, as the central human resource management team can identify departments with unused employees and relocate them to the impacted office. In contrast to the large literature arguing that draft-induced vacancies drew women into male-dominated workplaces (Goldin, 1991; Acemoglu et al., 2004), in our setting the inflow of women into the civil service was mostly associated with voluntary male exits amid an overheated wartime labor market rather than direct replacement of drafted workers.

2 Background

2.1 Institutional Details of Tokyo Civil Service

The Tokyo public service, comprising the *Tokyo-Fu* and *Tokyo-Shi*, managed public good provision across geographically distinct areas of Tokyo. Although these bodies operated over separate regions, their main offices were centrally located in downtown Tokyo.

In 1944, under intensifying war pressures and facing repeated military defeats, the central government mandated the amalgamation of *Tokyo-Fu* and *Tokyo-Shi* into a single entity, named *Tokyo-To*, through a federal bill. This merger, driven by operational inefficiencies and resource allocation challenges during the war, resulted in significant layoffs, particularly affecting *Tokyo-Shi* employees.

Tokyo-To employed approximately 15,000 highly educated staff, with 30 percent hired through national exams and the remainder locally. The organizational structure was hierarchical, dividing departments based on the public services provided, with some departments split into as many as four layers. Employees were assigned an occupation, such as engineers, managers, clerks, or assistants, along with a classification reflecting their professional expertise and experience. These assignments were determined upon entry into the organization, although transitions across occupations occasionally occurred. The recruitment process for non-assistant employees was merit-based, relying on competitive examinations and internal assessments. Within the civil service, workers were further categorized into upper and lower classes, with managers representing the upper class and clerks or engineers forming the lower class. Leadership positions, such as department heads, were typically filled by personnel recruited from the central government. For many employees, the typical career path involved progressing step by step up the job ladder, with clerks serving as the standard entry point for civil servants aiming to advance. However, the level of competition for promotions among employees was likely influenced by the conscription of a significant portion of the workforce into military service, which temporarily reduced the pool of available candidates. This shift may have also led to substantial changes in both the promotion process and the speed of advancement for positions such as clerks and managerial staff. In fact, many staff members were conscripted as soldiers, resulting in a severe manpower shortage for managing wartime administration (Tokyo Metropolitan Government, 2013).

2.2 Drafting Process of the Japanese Military Service

The Japanese military assembled its forces by calling active service troops and reservists, a process pivotal to our study's analysis of workforce dynamics during and post-conflict periods. The conscription system was designed to ensure a steady supply of personnel by mandating physical examinations for all eligible men upon reaching the age of 20, with those meeting the criteria randomly selected for service.

The drafting process was structured into several stages to efficiently classify conscripts based on their physical capabilities and the military's operational needs. Initially, all eligible men underwent a physical examination to determine their fitness for active duty. Those deemed fit were assigned to active service, where they served as soldiers for a fixed period. Upon completing active duty, these individuals transitioned to the reservist category, ensuring they remained available for military recall during national emergencies. For individuals who did not meet the physical standards for active duty, an alternative route, termed replacement service, was established. These men received training but were not immediately deployed, instead forming a reserve pool for supplementary support when required. Additionally, a subset of candidates was fully exempted from military obligations due to severe health conditions or other qualifying factors.

As the war intensified, particularly in the wake of Japan's challenges during the Second Sino-

Japanese War (1937–1945), the drafting system underwent significant modifications to address the military's increasing personnel demands. In 1943, reserve obligations were extended: the period of reserve service following active duty increased from 5–6 years to 15 years, and the secondary reserve period was prolonged until the age of 45. These changes ensured that the military could rely on a broader pool of trained personnel over an extended timeframe, providing critical support for wartime operations.

2.3 The Impact of the War on Tokyo-To

The war significantly disrupted the operations of *Tokyo-To* by removing a large share of its male workforce. The conscription system, which required eligible men to serve in the military, had profound implications for public institutions, especially in urban centers like Tokyo. Male workers in administrative, engineering, and clerical roles were not exempt from military obligations, leading to their systematic removal from public service positions as they were drafted into active or reserve duty. This mass conscription created substantial disruptions in the operations of public offices and left critical gaps in the workforce that urgently needed to be addressed.

In addition to conscription, the demands of wartime administration likely necessitated an increase in the number of staff employed by *Tokyo-To*. The rising responsibilities of the metropolitan government during the war made it essential to expand its workforce. Historical evidence suggests that both the city and the prefecture increased the number of employees and departments during this period (Tokyo Metropolitan Government, 2013).

2.4 Female Workers in the Tokyo Civil Service

Before the war, female employees were largely concentrated in specific departments and were almost absent from the core administrative divisions of the metropolitan government. Their roles were typically limited to auxiliary or supporting tasks, reflecting the gendered labor practices of the time. Units were generally structured to include a manager, clerks or engineers, and assistants. Female workers were primarily hired as assistants to support unit leaders in their tasks, often in offices with a lower concentration of engineers. Their continuation and promotion within the organization depended heavily on the approval of their respective unit managers, as well as on their working relationships with male colleagues—for example, men who were direct subordinates of the same unit manager or coworkers of clerks and engineers.

To compensate for the depletion of male workers, *Tokyo-To* rapidly increased the recruitment of women, who were not subject to military service. Women entered the public workforce in unprecedented numbers, particularly in assistant roles. Figure 4 documents this evolution. Panel (a) illustrates the sharp increase in the population of female public civil servants during World War II (WWII). The count of female employees rose significantly between 1938 and 1944, reflecting a direct response to the conscription of men. However, this trend reversed immediately after the war as male workers returned to their civilian roles, leading to a sharp decline in the number of female public civil servants.

Panel (b) provides further insight by examining the share of women in the public sector workforce during the same period. The proportion of female public civil servants also increased markedly during WWII, albeit remaining relatively small compared to the overall workforce. The rise in this share highlights not only the increasing reliance on women but also the gendered constraints of wartime employment policies, which often relegated women to lower-ranking and temporary roles within the organizational hierarchy. This unequal distribution persisted even after the war, as men resumed their pre-war positions, and women were often pushed out of public service.

3 Data

3.1 Structure of the personnel archive

The empirical work uses a worker–year panel built from annual staff directories of the Tokyo prefectural and municipal civil service. Each row corresponds to one person observed in one publication year, after line-level OCR records are collapsed to individuals. We retain only rows that pass automated name plausibility checks; preamble text, stamps, and index lines are excluded from the estimation sample but retained in the raw extract for auditing.

Identifiers and linkage. Each observation carries calendar year; an indicator for government level (Tokyo city, Tokyo prefecture, or Tokyo metropolis); a stable numeric identifier for the printed office; and a cross-year person identifier formed by exact and (within-office) fuzzy name matching. The organizational hierarchy read from the directories—bureau (*kyoku*), division (*bu*) when printed, section (*ka*), and unit (*kakari*)—is forward-filled from section headers so each worker is assigned to a full path in the tree; Figure 1 sketches the ordering used in Tokyo-To volumes after 1943. The printed job title is normalized to a standardized position label using a crosswalk table; unmatched strings are labeled unknown or passed through as raw text depending on the rule set.

Personnel fields. The extract includes the worker's name; gender from two Japanese-name heuristics (a conservative "legacy" rule and an augmented "modern" rule used in the main analysis); grade (civil-service grade, e.g. upper/lower tier, when printed); salary (monthly pay, typically in kanji numerals) and rank (court-rank strings such as senior/junior grades) parsed from the same text block as the name; and page or image coordinates for traceability. These salary and rank fields are present when the directory printed them for that line; coverage is incomplete in early years and degraded by OCR quality for some scans.

Military drafting. Some directory editions—notably under Tokyo prefecture during the war—explicitly flag employees who were conscripted. We record a draft indicator on those rows. When no such annotation appears for an edition, the indicator is missing or false. Treatment variables in the regressions aggregate *male* draftees to the office–position–year cell (see below).

Special pay (*rinji tokubetsu*). When the OCR text records a temporary special allowance, we retain that text; otherwise the allowance field is empty. Outcomes counting recipients sum indicators constructed from non-empty allowance records at the relevant level of aggregation.

Coverage. The merged master combines all published levels and years in the digitization project: Tokyo prefecture (1938–1941), Tokyo city (1937–1943), and Tokyo metropolis (1944 and selected postwar years through the 1950s), with gaps where volumes were not imaged or not processed. Figure 2 plots the number of unique names recovered per year, illustrating both the panel's coverage and the gaps in the source record. The time span used in the drafting tables is typically 1938–1945 for the wartime analysis; broader years enter descriptive and postwar exercises.

The two co-governing institutions (*Tokyo-Shi* and *Tokyo-Fu*) published separately until the 1943 merger; after that, records refer to the metropolitan government (*Tokyo-To*). The Tokyo-Fu volumes are the ones that list military draftees alongside continuing staff, which makes the draft shock measurable at the workplace level.

One consequence of the merger is that the mapping between *ka* (sections) and *kyoku* (bureaus) is not stable across all years: sections were reorganized or reassigned to different bureaus as part of the institutional consolidation. We can construct a reliable *kyoku*-level crosswalk using the broad bureau structure, but finer *ka*–*kyoku* linkages are not consistently traceable. This limits our ability to classify within-*ka* transfers precisely, and we therefore omit that channel from the transfer decomposition analysis (Table 5).

3.2 Secondary biographical volumes

A separate biographical series (not annual) covers higher-ranked Tokyo-Shi and Tokyo-To employees in certain years (e.g. 1942). It adds fields such as birth date, origin prefecture, education, career start date, and hobbies, and richer address information than the annual lists. We use these volumes only where noted in the appendix; they are not the basis for the main panel.

3.3 Digitization pipeline

The directories begin as page scans. We run the National Diet Library's neural OCR to produce XML with line text and coordinates, then Python routines sort columns, strip embedded salary and rank tokens, split names with Sudachi, infer office hierarchy, and apply the crosswalk and quality flags described above. Technical documentation of the pipeline appears in Appendix A.1. Recovery rates are lower for interwar script and scan quality than for postwar editions.

3.4 Construction of regression variables

The treatment variable is the count of drafted *male* workers in each office–position–year cell, with a log cumulative male baseline as a size control. Outcome variables include counts of incoming transfers (classified by organizational distance), new hires (by gender), an indicator for promotion (rank increase between consecutive years), retention (appearance in the following year's directory), bonus pay recipients, female share, and average tenure. We interact the treatment with rank buckets (rank 1: temporary/contract; rank 2: clerical/technical; rank 3: supervisory) and an engineer indicator. Baseline specifications include year, section (*ka*), and position fixed effects (with *kyoku* fixed effects in donor-office specifications), and standard errors cluster at the office level. Full construction details appear in Appendix A.5.

3.5 Final database and descriptive scale (Table 2)

The final analysis database is the merged, cleaned personnel master restricted to name-valid rows and enriched with the constructed variables above. Table 2 summarizes its scale; the full table appears with the empirical results, but we record the magnitudes here because they define the support for estimation.

Panel A counts unique persons and organizational units. In the full sample, there are 92,439 distinct workers nested in 119 bureaus (*kyoku*), 4,967 section identifiers (*ka* within *kyoku*), and 9,234 lowest-level units (*kakari*). 5,968 unique workers are classified as female under the modern gender heuristic. Worker–year counts by coarse job rank are 15,296 at rank 3 (supervisory *shiji*/*gishi*), 124,127 at rank 2 (clerical/technical *shoki*/*gite*), and 99,555 at rank 1 (temporary/contract *yatoi*/*shokutaku*); these categories overlap years and therefore do not sum to unique workers.

The "Drafted years" column narrows to publication years in which at least one worker is flagged as drafted (1938, 1940, 1941, 1943, 1944). There the sample contains 54,254 unique workers, 48 *kyoku*, 1,464 *ka*, 1,690 *kakari*, and 3,720 women, with 4,989, 74,352, and 29,312 rank-3/2/1 observations respectively. The 1944 column isolates the last wartime pre-surrender directory used in several cross-sections: 9,323 workers, 8 *kyoku*, 276 *ka*, 383 *kakari*, 702 women, and 470 / 8,013 / 789 observations by rank.

Panel B reports *kakari*-level means with standard deviations in parentheses. A typical unit has about 23 workers in the full sample (SD 102) versus 58 (SD 207) in drafted years and 24 (SD 39) in 1944; the dispersion reflects a long right tail of large offices. The average *kakari* lists 2.2 distinct position titles (SD 1.7), rising to 3.4 in drafted years and in 1944. Mean staffing per position slot within rank categories is reported separately for ranks 1–3 and shows the same pattern of heavier staffing in the drafted-year slice.

These counts underpin the regression samples: office–position–year cells aggregate workers within the hierarchy above, and the drafting subsamples retain substantial variation in both headcount and depth of the organization tree.

4 Empirical Model

4.1 Drafting specification

We estimate the causal effect of military drafting on office-level outcomes using the specifications below. The baseline unit of observation is position $\times$ kakari $\times$ year (1938–1945), unless noted. All specifications include year fixed effects and either ka (section) or kyoku (department) fixed effects, plus position fixed effects. Standard errors are clustered at the office level.

Incoming transfers (Tables 4 and 5). Count outcomes (transfers in) are modeled with OLS:

$$Y_{j,p,t} = \beta_1 D_{j,p,t} + \beta_2 \log(M_{j,p,t} + 1) + \gamma + \delta_p + \lambda_t + \varepsilon_{j,p,t}, \tag{1}$$

where $Y_{j,p,t}$ is the number of transfers into position p in office j in year t; $D_{j,p,t}$ is the number of male workers drafted from that position; and $M_{j,p,t}$ is the cumulative male stock. The coefficient β_1 measures the change in the expected number of transfers associated with one additional draft from the same position.

New hiring (Table 6). We use the same count specification with the number of new hires as

$Y_{j,p,t}$.

Short-run promotion (Table 8). Using a linear probability model, we regress a promotion indicator (rank increased from $t-1$ to t) on the number of drafts from the same office, for non-drafted workers observed in consecutive years. Fixed effects are year, ka, and position. We interact the treatment with gender, rank, and engineer indicators.

Bonus salary (Table 9). We regress the number of workers receiving *rinji tokubetsu* (temporary special) allowances on the number of drafts, in the 1944 cross-section. The unit of observation is position $\times$ office. Fixed effects are ka and position.

Retention (Table 10). We regress an indicator for whether a non-drafted worker appears in the next year's directory on the number of drafts from the same office. Fixed effects are year, ka, and position.

Donor offices (Table 11). We compare donor office–years (those with at least one outbound transfer) to non-donors at the ka $\times$ position $\times$ year level, with kyoku, position, and year fixed effects and log headcount where noted. Appendix Table 17 tests whether hiring differences predate transfers; Appendix Table 18 summarizes selection into transfer within donors.

Long-run outcomes of transferred workers (Table 12). We compare movers to destination and origin peers with kakari-, year-, and position-level fixed effects as in the table. Appendix Table 19 reports postwar rank and tenure for workers who remained in donor offices.

Gender composition and tenure (Table 7). Female share of workers and average tenure at the position $\times$ kakari $\times$ year level are modeled by OLS. The sample excludes *kanri* (higher-tier civil servant) positions where noted. Treatment is the count of drafted males from the same office–position. We control for cumulative male baseline and the same fixed effects as above.

Wartime draft shock and postwar female integration (Table 13). We regress postwar female outcomes in a *kakaricho*'s office on the wartime draft shock (share of males drafted) in that manager's wartime office, at the kakari $\times$ year level after 1947. Fixed effects condition on the kakaricho's wartime *ka* and wartime position; some specifications add postwar year fixed effects.

Postwar tenure (Table 15). For non-drafted workers observed in 1944, we regress postwar tenure (count of years appearing in civil service through 1955) on drafting exposure in their 1944 office. The unit of observation is the individual; treatment is any drafted or draft share; fixed effects are ka (or kyoku) and position as of 1944.

4.2 Identification strategy

Our estimand is the causal effect of military drafting on personnel outcomes, holding fixed the fine organizational and occupational structure in which a worker or office is observed. The identifying assumption is *conditional exogeneity*: conditional on the fixed effects included in each specification—typically year, section (*ka*), and position, and sometimes bureau (*kyoku*)—variation in the number of workers drafted from a position is orthogonal to unobserved shocks in the structural error term. Equivalently, potential outcomes for hiring, transfers, promotion, and retention do not differ systematically across draft counts once those fixed effects and baseline male counts are held constant.

This assumption would fail if personnel offices systematically assigned conscription risk or drafted workers on the basis of unobserved productivity shocks that also drive our outcomes. We mitigate concern about unobserved heterogeneity across large, coarse groups by including *ka* (section) fixed effects: Figure 1 illustrates how *ka* sits in the Tokyo-To hierarchy as the section at which substantive portfolios are organized, so comparisons are primarily within the same policy unit and job title. Year fixed effects absorb aggregate shocks to the draft and the wartime bureaucracy; position fixed effects absorb differences across standardized job titles (clerical, engineering, and so on). The draft count itself is determined outside the office by nationwide conscription eligibility (age, health, student or family exemptions), which supports treating residual variation as conditionally exogenous to local human-resource strategy once organizational cell and occupation are controlled for, and the pre-war balance evidence in Table 3 is consistent with this identifying assumption.

4.3 Balance of pre-war outcomes conditional on *ka* and occupation

Table 3 evaluates whether offices that later experienced drafting looked different *before* the war, using the pre-war panel (1934–1937, strictly before any military drafting began in 1938) restricted to units also observed in 1944. The unit of observation is *kakari* $\times$ position $\times$ year, as in the main regressions. Panel A compares unconditional means between cells with and without any drafting in 1944: drafted-in-1944 units display a much lower female share and a higher new-hire share, so simple cross-sectional comparisons confound drafting with baseline churn and gender mix.

Panel B addresses this by regressing each pre-war outcome on the 1944 male draft count (continuous, matching the treatment variable in our main regressions) while including *ka* fixed effects, position fixed effects, and year fixed effects—the same conditioning dimensions used in the main regressions—and controlling for log office headcount. After partialing out section, position, year, and size, the coefficient on future draft intensity is statistically indistinguishable from zero for every reported outcome: female share ($p = 0.15$), new-hire share ($p = 0.74$), promoted share ($p = 0.11$), external transfer share ($p = 0.70$), and headcount ($p = 0.18$). In sum, within section (*ka*) and position cells, pre-war trajectories look parallel across units that would later differ in draft intensity, supporting the conditional exogeneity assumption used for causal interpretation.

5 Results

Unless noted otherwise, regressions use the drafting specifications in the Empirical Model section: treatment is the count of male workers drafted from the same office–position (plus log cumulative male baseline where indicated), with year, ka (section), and position fixed effects, and standard errors clustered at the office level. Table 1 summarizes the unit of observation, treatment, and fixed effects for each table.

Table 1: Guide to Regression Tables

Table	Outcome	Unit of obs.	Fixed effects	Years
4, 5	Transfers in	pos × kakari × year	ka, pos, year	1938–45
6	New hires	pos × kakari × year	ka, pos, year	1938–45
7	Female share, tenure	pos × kakari × year	ka, pos, year	1938–45
8	Promoted (=1)	individual × year	ka, pos, year	1938–45
9	Bonus pay count	pos × office	ka, pos	1944
10	Retained (=1)	individual × year	ka, pos, year	1937–45
11	Office characteristics	ka × pos × year	kyoku, pos, year	1938–45
12	Career outcomes	individual	kakari, pos, year	all
13	Postwar females	kakari × year	wartime ka, pos, year	1947–
14	Female workers	office × year	year	1937–44

Notes: Treatment in all main drafting tables (4–10) is the count of males drafted from the same office–position, with log cumulative male baseline as a size control. Table 11 uses a donor indicator. Table 12 uses a transferred indicator with peer composition controls. Table 13 uses the wartime draft shock (share of males drafted from the kakaricho's wartime office). Standard errors are clustered at the office level throughout (kakari level in Table 12).

5.1 Transfers in and decomposition by distance

Table 4 estimates how incoming transfers from a different bureau (*kyoku*) group respond to local drafting; the sample is position × kakari × year (1938–1945). Column (1) reports a positive baseline slope of about 0.14 additional cross-bureau transfers per draft ($p < 0.05$). Column (2) adds rank interactions: the draft slope is larger for rank 3 (supervisory) positions and indistinguishable from zero for rank 1. Column (3) interacts with an engineer indicator (not precisely estimated). Column (4) interacts with the position's female employment share and yields a positive and significant interaction.

Table 5 decomposes incoming transfers into two channels by organizational distance: from a different *ka* within the same *kyoku* (columns 1–3); and from a different *kyoku* (columns 4–6). We omit within-*ka* transfers because *ka*–*kyoku* mappings cannot be reliably tracked across the 1943 Tokyo-Fu/Shi to Tokyo-To merger (see Section 3.1). The significant responses load on both remaining channels—about 0.29 per draft for same-*kyoku*, different-*ka* transfers (column 1) and about 0.14 for cross-*kyoku* transfers (column 4). The pattern implies that replacement workers typically cross section or bureau boundaries rather than reshuffling only adjacent teams, consistent with centralized redeployment.

5.2 New hiring

Table 6 estimates hiring responses at the position $\times$ kakari $\times$ year level. Columns (1)–(3) use total new hires: the draft count raises new hiring by about 0.22 per draft in the baseline (column 1, $p < 0.05$), with rank and engineer interactions in (2)–(3). Columns (4)–(6) decompose new female hires (all, rank 1 only, rank 2+): the overall female-hire slope is near zero and insignificant (column 4), while rank 2+ female hires tick up slightly (column 6). Columns (7)–(9) report analogous male counts: the total male slope is positive but noisy (column 7), whereas rank 2+ male hiring rises clearly (column 9). Male replacements therefore account for most of the intensive-margin hiring response; female hiring does not track draft counts at the office–position level. Comparing the transfer and hiring margins, each draft-induced exit generated about twice as many internal transfers (0.41) as external hires (0.22).

5.3 Gender composition and tenure

Table 7 uses the position $\times$ kakari $\times$ year panel (1938–1945). Columns (1)–(3) regress the female share of all workers on the draft count with optional rank and engineer interactions; coefficients are economically small and statistically insignificant, echoing the null for female hiring in Table 6. Draft-induced transfers and external hires were predominantly male, leaving no statistically significant change in office gender composition. Columns (4)–(6) use average worker tenure in the position as the outcome: the baseline draft coefficient is negative and significant (about -0.07 years per draft), implying shorter average seniority where drafting is more intense. Rank interactions in column (5) show an additional large negative slope for rank 3 positions. Engineer interactions in column (6) partially offset the tenure loss. High draft exposure therefore coincides with a younger, less tenured mix even as the female share is stable.

5.4 Short-run promotion

Table 8 studies promotions among non-drafted workers observed in the same office in consecutive years. The dependent variable is an indicator for a rank increase from $t-1$ to t; treatment is total males drafted from that office in year t. Column (1) reports a positive baseline effect of about 8.7 percentage points per additional draft ($p < 0.05$). Column (2) adds a female interaction: women receive an additional positive bump (about 2.8 points, $p < 0.01$). Column (3) allows rank-specific draft slopes; estimates are modest and less precisely separated. Column (4) shows a negative engineer interaction (about -4.2 points, $p < 0.01$), indicating smaller promotion responses in technical lines where grade rules may bind. Internal promotion therefore complements transfers and hiring, but not uniformly across demographics and occupations.

5.5 Bonus salary (*rinji tokubetsu*)

Table 9 uses the 1944 cross-section at the position $\times$ office level. The dependent variable is the count of non-drafted workers receiving temporary special (*rinji tokubetsu*) pay. Column (1) shows a positive draft slope (about 0.14 recipients per draft, $p < 0.05$). Column (2) interacts with rank: the baseline draft effect is strongest for rank 2 (regular civil servants), rank 1 shows a negative incremental slope, and rank 3 adds a positive incremental slope. Column (3) shows a positive engineer interaction: conditional on other terms, engineers are more likely to be placed on bonus schedules when drafts hit. Column (4) adds female-share interactions for treatment and level. The table supports using bonus pay as a margin of adjustment targeted to rank and technical roles.

5.6 Retention of non-drafted workers

Table 10 uses a linear probability model at the individual level: the dependent variable equals one if a non-drafted worker listed in year $t-1$ still appears in year t. The treatment is the number of males drafted from the same office–position in year t. Columns (1)–(4) add interactions with female (column 2), rank (3), and engineer (4). The point estimate on the draft count is about 0.9

percentage points and imprecise in every column; interactions are likewise small. We therefore do not find evidence that drafting systematically raised or lowered next-year survival among stayers once office–year–position structure is held fixed.

5.7 Donor offices

Table 11 compares *donor* office–years—those that sent at least one transfer out—to non-donors, at the ka $\times$ position $\times$ year level. Panel A reports unconditional differences: donors are more likely to record any new hire, record roughly 6 more hires on average, have slightly longer average tenure, and lower female share. Panel B adds kyoku, position, and year fixed effects plus log headcount: donors still exhibit higher probabilities and counts of new hiring, about 0.12 years longer tenure, and slightly lower female share. The pattern is consistent with offices tapped for outbound transfers having sufficient organizational slack—a stable core workforce combined with high entry-level flow—rather than being random slices of the hierarchy, though we cannot rule out that donor status also reflects unobserved office-level characteristics. Within donor offices, selection into transfer was not predicted by the worker's gender, tenure, or rank (Appendix Table 18). Appendix Table 17 shows that conditional on kyoku, position, year, and log headcount, offices that become donors in year t had about 5.1 additional new hires in year $t-1$ relative to non-donors (Panel B), suggesting that higher hiring levels in donor offices reflect pre-existing capacity rather than being solely a response to the outbound transfer. We interpret the donor analysis as descriptive: it characterizes the types of offices from which the central HR team drew transferees, but does not identify causal effects of donor status.

5.8 Long-run outcomes of transferred workers

Table 12 compares workers who move across offices to two peer groups, controlling for peer composition at the relevant office (average rank, average tenure, female share, and log headcount). Panel A contrasts transferees with coworkers already at the *destination* kakari in the transfer year (destination kakari, year, and position fixed effects). Transferred workers accumulate about 1.7

more years of post-transfer tenure, 10.5 percentage points higher postwar survival (any appearance 1947–1955), and 0.12 units greater rank advancement. Panel B contrasts the same transferees to colleagues who stayed at the *donor* kakari (donor kakari, year, position FE) and finds the same sign pattern with larger point differences: 3.4 additional years of tenure, 16.3 percentage points higher postwar survival, and 0.18 units greater rank gain. Appendix Table 19 shows that workers who remained in donor offices through the postwar period had lower standardized rank advancement than peers in same-*ka* offices without outbound transfers—about 0.31 standard deviations in the baseline specification.

These comparisons do not have a causal interpretation: transferred workers may differ from peers on unobserved dimensions despite the fixed effects and peer-composition controls. Appendix Table 18 shows that observable attributes—gender, tenure, and rank—do not predict selection into transfer within donor offices, but we cannot rule out selection on unobservables. We therefore interpret the outcome gaps as descriptive associations: workers who were moved tended to have better subsequent careers than comparable stayers, consistent with—but not proof of—organizational position mattering for career progression.

5.9 Wartime draft shock and postwar female integration

Although office-level drafting did not shift gender composition directly, it did increase hiring in low-ranked positions, potentially shaping the managerial experiences of workers promoted as a result of drafting. Table 13 tests the reduced-form relationship: does the wartime draft shock in a *kakaricho*'s (subsection chief) wartime office predict postwar female inclusion in that manager's postwar office?

The unit of observation is kakari $\times$ year (1947 onward), restricted to kakari-years whose kakaricho can be linked to a wartime office. The treatment variable is the mean share of males drafted from the kakaricho's wartime position $\times$ kakari $\times$ year cells. All specifications condition on the kakaricho's wartime *ka* (section) and wartime position fixed effects, so comparisons are within the same wartime organizational and occupational cell. Columns (1)–(2) use an indicator for any female worker; columns (3)–(4) use the count of females; columns (5)–(6) use the female share. Even-numbered columns add postwar year fixed effects.

The draft shock coefficient is small and statistically insignificant across all six specifications: point estimates range from -0.06 to $+0.02$ depending on the outcome, with standard errors that would accommodate effects of moderate size. Conditional on wartime section and position, the intensity of conscription in a manager's wartime office does not predict whether that manager's postwar office includes female workers. The null result is consistent with the finding in Tables 6 and 7 that office-level drafting did not alter gender composition during the war itself.

5.10 Voluntary male exits and female worker volume

Table 14 asks whether offices that lose more male workers through voluntary exits—departures unrelated to the draft—also employ more women. The unit of observation is office $\times$ year (1937–1944, $N \approx 3{,}031$). The treatment is the voluntary male exit rate: the share of male workers present in the office at $t-1$ who do not appear anywhere in the directory at t and were not drafted. Identification is cross-sectional within each year, with year fixed effects absorbing common time trends.

Column (1) reports a positive and significant baseline coefficient of about 2.9 additional female workers per unit of exit rate ($p < 0.05$): offices where a larger fraction of men departed voluntarily tend to employ more women that year. Columns (2) and (3) split the sample by government tier. The effect is larger and more precisely estimated for Tokyo-Fu (prefectural) offices (about 5.3, $p < 0.05$) than for Tokyo-Shi (city) offices (about 1.7, $p > 0.10$). We interpret this as descriptive evidence that voluntary male attrition creates organizational openings that are partially filled by female workers, consistent with a simple vacancy-chain mechanism. The cross-sectional design does not permit a causal interpretation—offices with high voluntary exit rates may differ from others on unobserved dimensions—but the positive association holds across specifications and is directionally consistent with the broader substitution patterns documented in Table 6.

5.11 Additional results

The Appendix reports additional results using the same identification strategy: postwar outcomes for the 1944 cohort (Table 15), headcount regressions (Table 16), lagged hiring tests for donor status (Table 17), characteristics of transferred workers (Table 18), and postwar outcomes for donor-office stayers (Table 19).

6 Conclusion

This paper documents how the Tokyo civil service replaced workers lost to military conscription during World War II. Using a novel personnel dataset covering 92,000 workers from 1928 to 1958, we trace the full chain of organizational responses—from immediate reallocation through postwar career outcomes.

Our findings reveal several key insights. First, under our conditional exogeneity design with pre-war balance checks, we estimate that offices responded to drafting through a combination of internal transfers (about 0.41 additional workers per draft, concentrated at the cross-section and cross-bureau level), new hiring (about 0.22 per draft, predominantly male), and internal promotion of remaining workers (8.7 percentage points per draft). The reliance on the cross-section margin rather than on within-team transfers, combined with the use of bonus salary allowances to retain scarce workers, points to a centralized human resource function managing the reallocation.

Second, the direct replacement of drafted men was overwhelmingly male: female hiring did not respond to office-level drafting, and the gender composition of affected offices remained unchanged. The aggregate wartime increase in female civil service employment was instead associated with the general expansion of wartime administration rather than the mechanical replacement of drafted workers. In our setting, the standard narrative from the U.S. literature—that wartime labor shortages directly drew women into male-dominated occupations—does not hold at the office level.

Third, transferred workers had better long-run career outcomes—higher retention, rank advancement, and tenure—than peers at both destination and origin offices. Observable attributes do not predict selection into transfer, but we cannot rule out unobserved selection, so these transfer-outcome comparisons are descriptive rather than causal. A reduced-form test finds no relationship between the wartime draft shock in a manager's office and postwar female inclusion, consistent with the null effect of drafting on wartime gender composition.

Our findings connect to work on coworker exits and internal labor markets (Jäger et al., 2025): we emphasize promotions and administrative allowances where private-sector studies emphasize wage spillovers, and we document cross-section reallocation (Table 5) when external hiring is costly. Relative to wartime labor-supply studies (Acemoglu et al., 2004; Goldin and Olivetti, 2013), office-level drafting does not move female shares (Tables 6 and 7) even though aggregate female civil-service employment rises (Figure 4), so direct replacement of drafted men was predominantly male. This complements evidence on the persistence of wartime female employment (Rose, 2018) while clarifying the margin—expansion of administration and general entry rather than draft-induced substitution. Salary schedules in the public sector limit wage flexibility (Goldin and Margo, 1992); Table 9 provides a rare administrative margin for adjustment.

By documenting the complete organizational response to conscription-driven labor supply shocks, this study contributes to the literatures on worker substitutability (Jäger et al., 2025) and wartime labor markets (Acemoglu et al., 2004; Goldin and Olivetti, 2013). Our results highlight that replacement dynamics in a bureaucracy with centralized personnel management differ markedly from dynamics in decentralized private-sector labor markets studied in prior work.

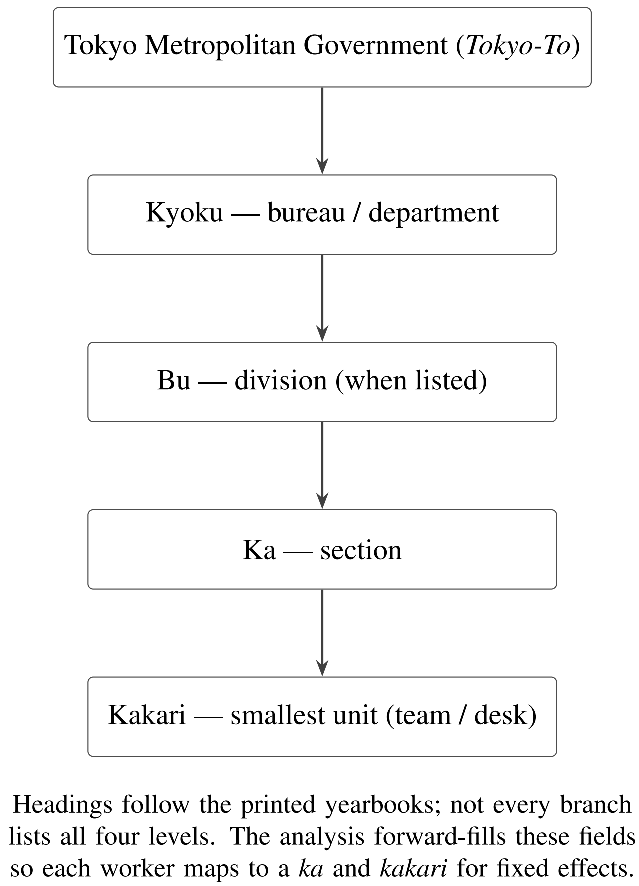


Figure 1: Administrative hierarchy in Tokyo metropolitan government (*Tokyo-To*) personnel directories. The diagram is schematic; some listings omit a *bu* line or combine labels.

Table 2: Summary Statistics

Panel A: Sample Counts

	Full Sample	Drafted Years	1944
Unique workers	92,439	54,254	9,323
Unique kyoku	119	48	8
Unique ka (ka × kyoku)	4,967	1,464	276
Unique kakari (kakari × ka × kyoku)	9,234	1,690	383
Unique women	5,968	3,720	702
Workers at rank 3 (shiji/gishi)	15,296	4,989	470
Workers at rank 2 (shoki/gite)	124,127	74,352	8,013
Workers at rank 1 (yatoi/shokutaku)	99,555	29,312	789

Panel B: Kakari-Level Averages

	Full Sample	Drafted Years	1944
Workers per kakari	22.7 (102.4)	58.3 (206.6)	24.3 (38.7)
Unique positions per kakari	2.2 (1.7)	3.4 (2.6)	3.4 (2.3)
Workers per rank-3 role	3.5 (22.5)	5.3 (47.2)	2.3 (3.8)
Workers per rank-2 role	12.8 (48.5)	19.3 (66.1)	8.5 (14.7)
Workers per rank-1 role	11.5 (36.1)	23.6 (67.4)	6.3 (8.3)

Notes: Panel A reports unique counts for the full sample, the subset of years with any drafted workers (1938, 1940, 1941, 1943, 1944), and 1944 alone. Rank 3 = *shiji*/*gishi* (supervisory); Rank 2 = *shoki*/*gite* (clerical/technical staff); Rank 1 = *yatoi*/*shokutaku* (temporary/contract). Panel B reports kakari-level averages (standard deviations in parentheses). A kakari is the lowest organizational unit (team), nested within ka (section) and kyoku (bureau).

Table 3: Balance Test: Pre-War Characteristics (1934–1937) by Future Drafting Status (1944)

Panel A: Unconditional Comparison

	Drafted in 1944		Not Drafted			
	Mean	SD	Mean	SD	Diff	p-value
Female Share	0.014	0.075	0.027	0.089	-0.013	0.373
No. New Hires	5.436	23.292	9.846	33.547	-4.410	0.422
No. Promoted	0.330	2.205	0.000	0.000	0.330	<0.001
No. Internal Transfers	91.293	233.354	110.872	203.717	-19.579	0.565
N Workers	104.704	249.954	127.795	228.190	-23.091	0.543
Avg Rank	2.146	0.795	2.103	0.680	0.043	0.704
Offices	22		8			
Obs (kakari × pos × year)	734		39			

Panel B: Conditional Comparison (Ka + Position + Year FE)

	Coef	SE	t-stat	p-value	N
Female Share	-0.0002	0.0001	-1.50	0.152	571
No. New Hires	-0.0987	0.1263	-0.78	0.446	571
No. Promoted	0.0132	0.0087	1.53	0.146	571
No. Internal Transfers	1.1205**	0.4533	2.47	0.025	571
N Workers	1.1397	0.8141	1.40	0.181	571
Avg Rank	0.0000	0.0000	0.00	1.000	571

Panel C: Pre-War Promotion by Future Donor Status (Ka + Occupation + Year FE)

	Coef	SE	t-stat	p-value	N
No. promoted workers	0.0042***	0.0009	4.88	<0.001	159
Future donor offices	15 offices				
Non-donor offices	1 offices				

Notes: Sample restricted to offices observable in both the pre-war period (1934–1937, before any military drafting) and in 1944. Unit of observation: kakari × position × year, matching the regression panel. Panel A reports unconditional means split by whether the office experienced any military drafting in 1944. Panel B reports coefficients from OLS regressions of each pre-war characteristic on the 1944 male draft count (continuous, matching the treatment variable in the main regressions), with section, position, and year fixed effects, controlling for log office headcount (except when outcome is headcount). Transfer and hiring outcomes are entered in raw counts (internal transfers and new hires, not shares). Standard errors clustered at the office level (30 clusters). Panel C reports the coefficient from an OLS regression of pre-war number of promoted workers (at the office × occupation × year level) on a future donor indicator (offices that sent at least one worker to a drafted office during 1938–1944), with section, occupation, and year fixed effects, clustered at the office level (16 clusters). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 4: Worker Reallocation: Transfers In

Model:	Baseline (1)	× Rank (2)	× Eng. (3)	× Gender (4)
Variables				
No. drafted workers	0.1438**	0.1333***	0.1991	0.1716***
	(0.0600)	(0.0364)	(0.1461)	(0.0617)
No. drafted × Rank 1		0.1316		
		(0.2609)		
No. drafted × Rank 3		0.8801**		
		(0.4214)		
No. drafted × Engineer			-0.0778	
			(0.1378)	
Fixed-effects				
year_num	Yes	Yes	Yes	Yes
ka_id	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
Fit statistics				
Observations	7,409	7,409	7,409	7,409
R^2	0.33206	0.33211	0.33209	0.33218
No. clusters	1,192	1,192	1,192	1,192

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: position × kakari × year (1938–1945). Dependent variable: number of workers who transferred in from a different bureau group. Transfers are defined as workers whose normalized bureau (kyoku) group changed from the prior year, using a crosswalk that accounts for the 1943 Tokyo-Fu/Shi to Tokyo-To merger. Column (2) interacts the treatment with rank indicators (rank 1 = *yatoi*/*shokutaku*; rank 3 = *shuji*/*gishi*; rank 2 is the omitted category). Column (3) interacts with an engineer indicator. Column (4) interacts with the position's female employment share. Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 5: Transfer Decomposition by Organizational Distance

	Retention (same section)		Close transfer (same bureau)		Distant transfer (diff. bureau)	
Model:	Base (1)	× Eng. (2)	Base (3)	× Eng. (4)	Base (5)	× Eng. (6)
Variables						
No. drafted workers	0.2269***	0.1408	0.1001**	0.0869	0.0809***	0.0575
	(0.0730)	(0.1157)	(0.0403)	(0.0919)	(0.0154)	(0.0362)
No. drafted × Engineer		0.1213		0.0186		0.0331
		(0.1236)		(0.0957)		(0.0381)
Fixed-effects						
year_num	Yes	Yes	Yes	Yes	Yes	Yes
ka_id	Yes	Yes	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	7,409	7,409	7,409	7,409	7,409	7,409
R^2	0.28608	0.28611	0.28497	0.28497	0.30613	0.30615
Size of the 'effective' sample	1,192	1,192	1,192	1,192	1,192	1,192

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: position × kakari × year (1938–1945). "Retention" (columns 1–2): workers remaining in the same ka group (same bureau and section). "Close transfer" (columns 3–4): workers from the same kyoku group but a different ka group. "Distant transfer" (columns 5–6): workers from a different kyoku group. Ka groups are constructed from worker flows across consecutive years: two (kyoku, ka) cells in adjacent years are linked if ≥40% of workers in the source cell appear in the destination cell. Kyoku groups use a crosswalk that accounts for the 1943 Tokyo-Fu/Shi to Tokyo-To merger. Engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 6: New Hiring in Response to Military Drafting

	No. new hires			No. new female			No. new male		
Model:	Baseline (1)	× Rank (2)	× Eng. (3)	All (4)	Rank 1 (5)	Rank 2+ (6)	All (7)	Rank 1 (8)	Rank 2+ (9)
Variables									
No. drafted workers	0.2057**	0.2471***	0.0967	0.0078	-0.0050	0.0128**	0.2046	-0.0731*	0.2777***
	(0.0998)	(0.0914)	(0.0863)	(0.0067)	(0.0046)	(0.0056)	(0.1319)	(0.0429)	(0.1035)
No. drafted × Rank 1		-0.4473***							
		(0.1353)							
No. drafted × Rank 3		1.006*							
		(0.5752)							
No. drafted × Engineer			0.1536						
			(0.1366)						
Fixed-effects									
year_num	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ka_id	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics									
Observations	7,409	7,409	7,409	7,409	7,409	7,409	7,409	7,409	7,409
R^2	0.24446	0.24492	0.24458	0.18309	0.15617	0.19213	0.38322	0.25380	0.36684
No. clusters	1,192	1,192	1,192	1,192	1,192	1,192	1,192	1,192	1,192

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: position × kakari × year (1938–1945). Columns (1)–(3): dependent variable is total new hires. Columns (4)–(6): new female hires (all, rank 1, rank 2+). Columns (7)–(9): new male hires (all, rank 1, rank 2+). Column (2) interacts the treatment with rank indicators (rank 1 = *yatoi*/*shokutaku*; rank 3 = *shuji*/*gishi*; rank 2 is the omitted category). Column (3) interacts with an engineer indicator. Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 7: Gender Composition and Tenure

Dep. var:	Female share			Avg. tenure		
	Baseline	× Rank	× Eng.	Baseline	× Rank	× Eng.
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
No. drafted workers	0.0006	0.0005	0.0026	-0.0677***	-0.0668***	-0.1091***
	(0.0009)	(0.0009)	(0.0021)	(0.0210)	(0.0215)	(0.0385)
No. drafted × Rank 1		0.0016			0.0106	
		(0.0026)			(0.0464)	
No. drafted × Rank 3		-0.0123			-0.5821*	
		(0.0132)			(0.3177)	
No. drafted × Engineer			-0.0029			0.0605
			(0.0022)			(0.0392)
Fixed-effects						
year_num	Yes	Yes	Yes	Yes	Yes	Yes
Ka	Yes	Yes	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	6,593	6,593	6,593	6,593	6,593	6,593
R^2	0.42749	0.42757	0.42762	0.56720	0.56743	0.56730
No. clusters	1,187	1,187	1,187	1,187	1,187	1,187

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions at the position × kakari × year level (1938–1945). Columns (1)–(3): dependent variable is female share of all workers. Columns (4)–(6): dependent variable is average tenure (years in data). Column (1) and (4) report baseline effects. Columns (2) and (5) interact draft exposure with rank indicators (rank 1 = *yatoi*/temporary; rank 3+ = senior; rank 2 = base category, absorbed by position FE). Columns (3) and (6) interact draft exposure with an engineer indicator (positions with an engineering title). All specifications include log(cumulative male baseline + 1) as a control (not displayed), as well as year, ka, and position fixed effects. Standard errors clustered at the office level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 8: Promotion Response to Drafting: Team × Occupation Level

Dep. var: No. promoted workers (team × occupation × year)			
	Baseline	× Rank	× Engineer
Model:	(1)	(2)	(3)
Variables			
No. drafted (office)	0.0267**	0.0231**	0.0262**
	(0.0107)	(0.0114)	(0.0117)
No. drafted × Rank 1 share		0.0154	
		(0.0112)	
No. drafted × Engineer			0.0013
			(0.0147)
Fixed-effects			
year_num	Yes	Yes	Yes
ka_id	Yes	Yes	Yes
occupation	Yes	Yes	Yes
Fit statistics			
Observations	2,770	2,770	2,770
R^2	0.28819	0.28825	0.28819
Size of the 'effective' sample	961	961	961

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: team (kakari) × occupation × year. Outcome: number of promoted workers in the cell (rank increased from $t-1$ to t). Sample: non-drafted workers present in the same team in consecutive years. Treatment: number of male workers drafted from the same team. Columns (2) interacts with the cell-level share of rank-1 workers; Column (3) interacts with an engineer indicator. All specifications include year, section (ka), and occupation fixed effects, and control for log cumulative male baseline. Standard errors clustered at the team level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 9: Bonus Salary (*Rinji Tokubetsu*) Response to Drafting

Dep. var: No. workers receiving rinji tokubetsu

Model:	Baseline (1)	$\times$ Rank (2)	$\times$ Engineer (3)	$\times$ Gender (4)
Variables				
No. drafted $\times$ Female share				-0.4321
				(0.3896)
No. drafted workers	0.1367**	0.1550***	0.0224	0.1564**
	(0.0529)	(0.0587)	(0.0424)	(0.0697)
No. drafted $\times$ Rank 1		-0.1854***		
		(0.0610)		
No. drafted $\times$ Engineer			0.1494*	
			(0.0758)	
Female share				1.787*
				(0.9378)
Fixed-effects				
ka_id	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
Fit statistics				
Observations	576	576	576	576
R^2	0.45717	0.46261	0.46430	0.46223
Size of the 'effective' sample	161	161	161	161

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regression using 1944 cross-section. Unit: position $\times$ office. Dependent variable: number of non-drafted workers receiving *rinji tokubetsu* (temporary special allowance). Treatment: number of male workers drafted from the same office $\times$ position. Column (1) is the baseline specification. Column (2) interacts the treatment with rank indicators (Rank 2 = regular civil servant is the base; Rank 1 = *yatoi*/*shokutaku*; Rank 3 = *shuji*/*gishi*). Column (3) interacts with an engineer indicator. Column (4) interacts with the female share of workers in the position. All specifications include ka and position fixed effects and control for log cumulative male baseline. Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 10: Retention of Non-Drafted Workers

Dep. var: Retained (=1 if worker appears in year t)				
	Baseline	× Gender	× Rank	× Engineer
Model:	(1)	(2)	(3)	(4)
Variables				
No. drafted workers	0.0091	0.0091	0.0130	0.0130
	(0.0062)	(0.0058)	(0.0098)	(0.0085)
No. drafted × Female		0.0022		
		(0.0138)		
No. drafted × Rank 1			-0.0090	
			(0.0091)	
No. drafted × Engineer				-0.0123
				(0.0090)
Female		-0.0082		
		(0.0060)		
Fixed-effects				
year_num	Yes	Yes	Yes	Yes
Ka	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
Fit statistics				
Observations	190,187	190,187	190,187	190,187
R^2	0.33710	0.33711	0.33711	0.33711
No. clusters	1,449	1,449	1,449	1,449

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS linear probability model. Dependent variable: retained (=1 if worker appears in year t). Treatment: number of male workers drafted from the same office × position in year t. All four columns use the full sample of non-drafted workers observed in any year from 1937 to $t-1$, using their most recent observation for office and position assignment. Column 1 estimates the baseline effect. Column 2 interacts the treatment with a female indicator. Column 3 interacts with rank dummies (Rank 1 = low-rank/temporary; Rank 3 = high-rank; Rank 2 is the omitted category). Column 4 interacts with an engineer-position indicator. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 11: Characteristics of Donor Offices (Offices with Transfers Out)

Panel A: Unconditional Comparison

	Donor Mean	Donor SD	Non-Donor Mean	Non-Donor SD	Diff	p-value
No. New Hires	9.251	18.899	3.317	9.719	5.934	<0.001
Avg Tenure (years)	2.880	1.301	2.636	1.516	0.245	<0.001
Female Share	0.026	0.079	0.044	0.142	-0.019	<0.001
No. Promoted	2.131	5.492	0.799	4.102	1.332	<0.001
No. Workers	18.825	30.647	12.144	27.226	6.682	<0.001

Panel B: Conditional Comparison (Position × Year FE)

	Office FE	Dept. FE
No. New Hires	2.6272***	3.9355**
	(0.8860)	(1.7100)
Avg Tenure (years)	0.0723**	0.1218***
	(0.0359)	(0.0364)
Female Share	-0.0090**	-0.0079**
	(0.0043)	(0.0035)
No. Promoted	0.3633	0.7532**
	(0.2532)	(0.3523)
No. Workers	4.0211**	3.7014*
	(1.6078)	(2.0093)
Observations	6,507	6,603

Notes: Sample: ka × position × year (1938–1945). A *donor* office is one that had at least one worker transfer out in that year. Panel A reports unconditional means. Panel B reports coefficients from OLS regressions on a donor indicator, with position and year fixed effects plus either office (ka) or department (kyoku) fixed effects, controlling for log office headcount. Standard errors clustered at the office level (1197 clusters). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 12: Long-Run Outcomes of Transferred Workers

Panel A: Transferred vs. Destination Office Peers

Model:	Tenure After (1)	Postwar Surv. (2)	Postwar Yrs (3)	Rank Gain (4)
Variables				
Transferred (=1)	1.686***	0.1054***	0.3158***	0.1214***
	(0.0791)	(0.0057)	(0.0182)	(0.0070)
peer_avg_rank	-0.3034	-0.0196	-0.0947	-0.0902**
	(0.3673)	(0.0238)	(0.0699)	(0.0378)
peer_avg_tenure	-0.1895***	-0.0064*	0.0008	0.0039
	(0.0521)	(0.0035)	(0.0101)	(0.0053)
peer_female_share	-2.252	-0.2035	-0.2974	-0.0574
	(2.128)	(0.1531)	(0.4545)	(0.1672)
log(peer_headcount+1)	-0.0840	0.0066	0.0082	-0.0125
	(0.0991)	(0.0073)	(0.0230)	(0.0088)
Fixed-effects				
Dest. Kakari	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Position	Yes	Yes	Yes	Yes
Fit statistics				
Observations	75,165	75,165	75,165	75,165
R^2	0.07759	0.06469	0.06891	0.31177
No. clusters	860	860	860	860

Clustered (Dest. Kakari) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Panel B: Transferred vs. Donor Office Peers

Model:	Tenure After (1)	Postwar Surv. (2)	Postwar Yrs (3)	Rank Gain (4)
Variables				
Transferred (=1)	3.361***	0.1634***	0.4800***	0.1798***
	(0.0944)	(0.0051)	(0.0172)	(0.0081)
peer_avg_rank	-0.4870	-0.0311	-0.1597**	-0.1020***
	(0.4083)	(0.0244)	(0.0778)	(0.0341)
peer_avg_tenure	-0.1685**	-0.0064	0.0003	0.0038
	(0.0669)	(0.0042)	(0.0137)	(0.0050)
peer_female_share	-2.715	-0.2503	-0.6472	-0.0573
	(2.403)	(0.1824)	(0.6162)	(0.1809)
log(peer_headcount+1)	-0.1151	-0.0028	-0.0269	-0.0139
	(0.1227)	(0.0076)	(0.0265)	(0.0089)
Fixed-effects				
Donor Kakari	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Position	Yes	Yes	Yes	Yes
Fit statistics				
Observations	76,723	76,723	76,723	76,723
R^2	0.10984	0.07574	0.07826	0.33013
No. clusters	944	944	944	944

Clustered (Donor Kakari) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions comparing transferred workers to peers (individual-level data). Panel A uses destination-office peers in the transfer year; Panel B uses donor-office peers in the year before transfer. Both panels control for peer composition and include the fixed effects listed in the table. Standard errors are clustered at the kakari level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 13: Reduced Form: Wartime Draft Shock and Postwar Female Integration

	Any female (=1)		N females		Female share	
Model	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Draft shock	-0.0536	-0.0564	-0.4470	-0.3880	0.0202	0.0183
	(0.1492)	(0.1490)	(0.3951)	(0.3702)	(0.0499)	(0.0503)
Kakari size	0.0081***	0.0081***	0.0537***	0.0541***	0.0001*	0.0001
	(0.0010)	(0.0009)	(0.0095)	(0.0092)	(6.3×10^{-5})	(6.69×10^{-5})
Fixed-effects						
wt_ka	Yes	Yes	Yes	Yes	Yes	Yes
wt_pos	Yes	Yes	Yes	Yes	Yes	Yes
year_num		Yes		Yes		Yes
Fit statistics						
Observations	2,686	2,685	2,686	2,685	2,686	2,685
R^2	0.25972	0.26101	0.57891	0.58173	0.15465	0.15632
No. clusters	1,322	1,322	1,322	1,322	1,322	1,322

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: kakari $\times$ year (1947 onward). The sample includes postwar kakari-years whose *kakaricho* (subsection chief) can be linked to a wartime office. The treatment variable is the mean share of males drafted from the kakaricho's wartime position $\times$ kakari $\times$ year cells. Fixed effects condition on the kakaricho's wartime *ka* (section) and wartime position; even-numbered columns add postwar year fixed effects. Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 14: Voluntary Male Exits and Female Worker Volume

Dependent Variable:	n_female_t		
	Full	Tokyo-Fu	Tokyo-Shi
Model:	(1)	(2)	(3)
Variables			
Voluntary exit rate	2.910**	5.334***	1.664
	(1.088)	(1.690)	(1.362)
Fixed-effects			
year_num	Yes	Yes	Yes
Fit statistics			
Observations	3,031	1,205	1,641
R^2	0.01751	0.02974	0.00743

Clustered (kyoku_norm) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: office $\times$ year (1937–1944). Dependent variable is the total number of female workers in the office. Treatment is the voluntary male exit rate: the share of male workers present in the office at $t-1$ who do not appear anywhere in the directory at t and were not drafted. Offices with no $t-1$ presence receive an exit rate of zero. All specifications include year fixed effects. Column (1) uses the full sample; columns (2) and (3) restrict to Tokyo-Fu (prefectural government) and Tokyo-Shi (city government) offices respectively. Standard errors clustered at the bureau (kyoku) level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

References

Daron Acemoglu, David H. Autor, and David Lyle. Women, war, and wages: The effect of female labor supply on the wage structure at midcentury. *Journal of Political Economy*, 112(3):497–551, 2004. doi: 10.1086/383100. URL https://doi.org/10.1086/383100.

Andrew Garin. Do place-based industrial interventions help "left-behind" workers? lessons from WWII and beyond. NBER Working Paper 33418, National Bureau of Economic Research, January 2025.

Claudia Goldin. The role of world war II in the rise of women's employment. *American Economic Review*, 81(2):741–756, 1991.

Claudia Goldin. The quiet revolution that transformed women's employment, education, and family. *American Economic Review*, 96(2):1–21, May 2006. doi: 10.1257/000282806777212350. URL https://www.aeaweb.org/articles?id=10.1257/000282806777212350.

Claudia Goldin and Robert A. Margo. The great compression: The wage structure in the united states at mid-century. *The Quarterly Journal of Economics*, 107(1):1–34, 1992. doi: 10.2307/2118327.

Claudia Goldin and Claudia Olivetti. Shocking labor supply: A reassessment of the role of world war ii on women's labor supply. *American Economic Review*, 103(3):257–262, 2013.

Simon Jäger, Jörg Heining, and Nathan Lazarus. How substitutable are workers? evidence from worker deaths. *American Economic Review, conditionally accepted*, 2025.

Walter Y. Oi. Labor as a quasi-fixed factor. *Journal of Political Economy*, 70(6):538–555, 1962. doi: 10.1086/258715. URL `https://doi.org/10.1086/258715`.

Evan K. Rose. The rise and fall of female labor force participation during world war ii in the united states. *The Journal of Economic History*, 78(3):673–711, 2018. doi: 10.1017/S0022050718000323.

Tokyo Metropolitan Government. Compilation of historical materials of Tokyo II: Establishment of the Tokyo metropolitan system (vol. 1), 2013.

A Appendix

A.1 National Diet Library OCR stack

Scanned directory pages are processed with the National Diet Library (NDL) open-source document OCR (version 2.x), which chains several neural and classical steps before line recognition. The pipeline, controlled through a configuration file and a command-line driver, typically runs in four stages: (i) *page separation*, which detects and splits two-up spreads when present; (ii) *deskew*, which corrects small rotation; (iii) *layout extraction*, which detects text regions on the page with a convolutional detector; and (iv) *line OCR*, which transcribes each line with a Japanese-character recognizer trained on historical print. The system can emit reading-order information, ruby (furigana) handling, and optional dumps of intermediate crops. For archival batch jobs we run inference on GPU nodes and request XML output so that each line retains its text content together with coordinates (and page or image identifiers) for downstream parsing.

The implementation bundles pretrained weights for separation, layout, and recognition; line recognition uses a character inventory suited to kanji–kana mixtures found in prewar administrative volumes. Batch runs store images alongside machine-readable XML in a dedicated output folder. Image quality and script style vary by era: interwar calligraphy and wartime paper can reduce accuracy relative to postwar editions, which motivates explicit validation and noise flags in the compilation stage described below.

A.2 From OCR XML to row-level personnel records

The extraction stage treats NDL XML as the interchange format between OCR and statistical analysis. For each publication year and government level (Tokyo city, Tokyo prefecture, or Tokyo metropolis), routines ingest all XML files for that wave. For each page they parse the XML tree, extract every text line with its bounding box, and cluster lines into vertical columns so that reading order follows traditional right-to-left columns and top-to-bottom order within a column—matching how names and titles appear in the printed directories.

Before splitting names, the code strips recurring metadata embedded in the same string as personal names: monthly salary expressed in kanji numerals, court rank strings, and grade or appointment markers. Remaining text is tokenized with SudachiPy (mode tuned for conservative splits). Tokens tagged as person names are retained; other material is kept for position matching and diagnostics. Job titles are matched against an external position crosswalk that maps variant strings to standardized labels used in analysis; unmatched titles are labeled as unknown or passed through as raw text depending on the rule set.

A.3 Compilation: offices, positions, and noise flags

A second stage converts the raw line-level table into an analysis-ready file for that year. It classifies lines as office headers using suffix patterns (e.g., bureau, division, section), entries in the crosswalk, and guards against confusing a position title (e.g., section chief) with an office name. Detected headers are forward-filled so that each employee row carries a nested administrative hierarchy (bureau through subsection, with normalized orthography mapping from historical to modern characters where applicable). The routine flags table-of-contents or index pages, library stamps, and other non-employee lines, and sets a boolean indicating whether a row plausibly represents a person's name (length, character set, particle checks). Heuristic gender labels are assigned from historical Japanese naming patterns (including conservative and extended rule sets). Drafting and wartime status markers are applied where encoded in the source text.

A.4 Master panel and identifiers

After each year-level file is produced, we concatenate all editions into a long panel. Each distinct office string receives a deterministic integer identifier for use in fixed-effects specifications. Cross-year person identifiers are assigned on name rows by grouping identical normalized names and, where implemented, applying similarity thresholds within offices to merge likely duplicate transcriptions; short names may be restricted to exact matching to limit false merges. Page number, image filename, and line coordinates from OCR are retained for auditing and for linking back to scans.

Implementation details, software dependencies, and data documentation are available in the supplementary materials; large model weights and proprietary crosswalks may reside on secured storage.

A.5 Construction of regression variables (details)

This subsection maps the master panel to the dependent and independent variables used in the Empirical Model and in the results tables. Unless noted, position enters as whitespace-stripped titles, and year enters as calendar year. Standard errors cluster at the office identifier.

Treatment (draft exposure). For each office j, normalized position p, and year t, let D_{jpt} denote the count of drafted *male* workers attached to that office–position in t. Let M_{jpt} denote the count of distinct male workers observed in (j, p) at any date strictly before t; we control for $\log(M_{jpt} + 1)$ as a size measure. Female workers are excluded from D_{jpt} so the treatment reflects conscription of men.

Transfers and new hires. Sorting workers by person identifier and year, we compare each worker's current office (and *ka*, *kyoku*, *kakari*) to the previous year. An incoming transfer is a worker present in year t whose prior-year office differs; mutually exclusive counts classify transfers by whether the origin shares the same *ka* and *kyoku*, shares *kyoku* only, or lies in a different *kyoku*, accounting for the 1943 administrative merger via a bureau crosswalk. A new hire is a worker who appears for the first time in year t, excluding the first year a newly opened office appears in the data. Female and male new hires are split using the modern gender classification.

Rank buckets for interactions. Positions are mapped to three ranks for heterogeneity: rank 1 (temporary/contract titles such as *yatoi*/*shokutaku*), rank 3 (supervisory or designated senior titles, with rules that depend on whether the year is before or after 1948), and rank 2 (all other classified titles). The engineer indicator flags positions whose normalized title contains the engineering marker used in the administrative job list.

Promotion. For non-draft workers observed in consecutive years in the same office, promotion is an indicator that the discrete rank category increases from $t - 1$ to t.

Retention. For non-draft workers, retention is an indicator that the person appears again in year t conditional on appearing in $t - 1$, with office–position assignment taken from the directory.

Bonus pay (*rinji*). The dependent variable is the count of workers in (j, p) with a recorded *rinji tokubetsu* allowance in the 1944 cross-section (or the year specified in the table).

Composition and tenure. At the position $\times$ kakari $\times$ year level, female share is the share of workers classified as female under the modern gender rule; average tenure is mean years since first appearance in the panel, computed within the relevant sample restrictions.

Fixed effects. We form section fixed effects by combining bureau and section labels so that effects nest within bureau where both are observed. Year, section, and position fixed effects match the tables in the paper; some specifications use bureau or office identifiers as noted in each table's notes.

Figure 2: Number of unique names by year

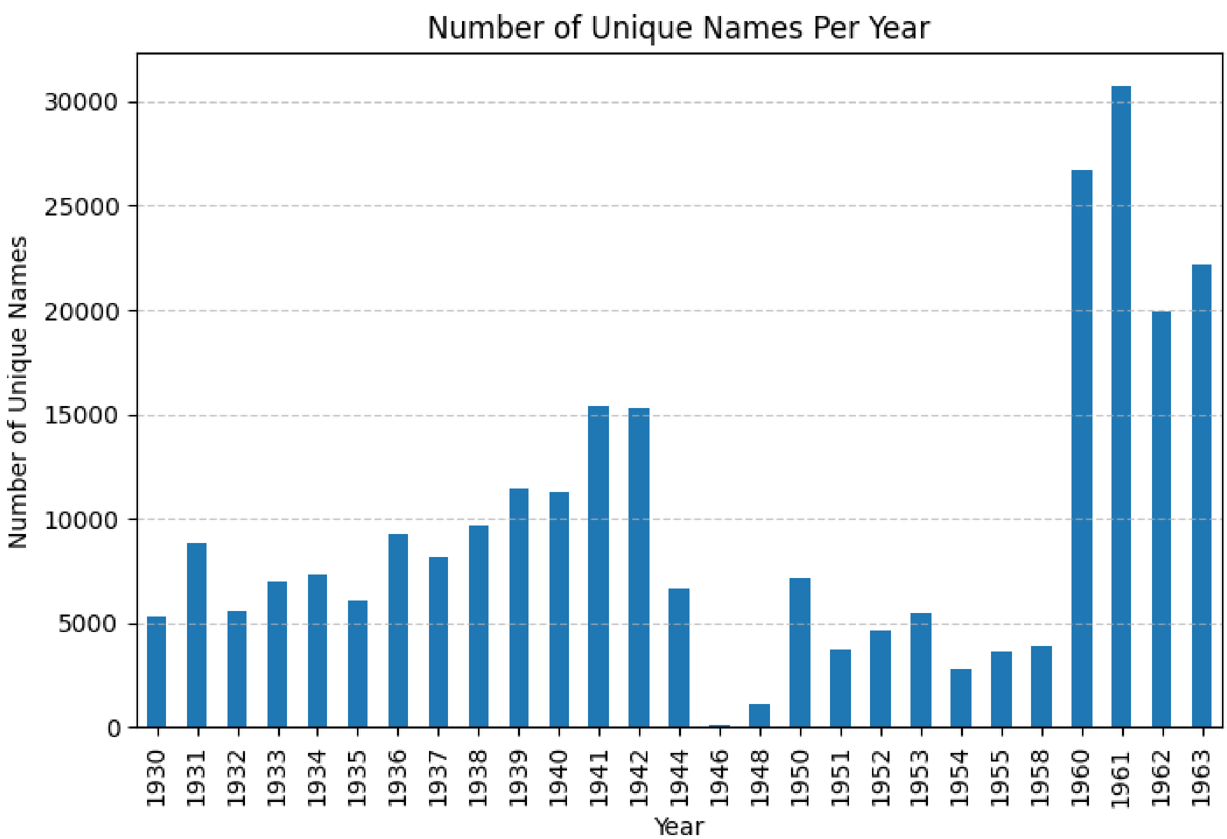


Table 15: Postwar Outcomes of 1944 Colleagues (1946–1955)

Dependent Variables:	rank_z_change		avg_female_share_dest	
	Promotion		Dest. F.Share	
Model:	(1)	(2)	(3)	(4)
Variables				
Draft share	-0.0919	-0.0929	-0.0018	-0.0018
	(0.1438)	(0.1457)	(0.0070)	(0.0059)
Female (=1)		-0.2087		0.1283***
		(0.1482)		(0.0477)
Draft share × Female		0.0047		0.0310
		(0.7866)		(0.1611)
Fixed-effects				
ka_id_1944	Yes	Yes	Yes	Yes
pos_norm_1944	Yes	Yes	Yes	Yes
Fit statistics				
Observations	1,281	1,281	1,281	1,281
R^2	0.46085	0.46163	0.09966	0.24099

Clustered (office_id_1944) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Sample: non-drafted workers observed in 1944. Columns (1)–(2): dependent variable is z-scored rank change (postwar avg. minus 1944 rank, standardized within era). Column (2) includes gender interaction. Columns (3)–(4): dependent variable is average female share of the worker's office-position across postwar years. Column (4) includes gender interaction. All specifications include ka and position fixed effects from 1944 and control for log cumulative male stock. Standard errors clustered at the office level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 16: Effect of Military Drafting on Remaining Office Size

Model:	Baseline (1)	× Rank (2)	× Eng. (3)	× Gender (4)
Variables				
No. drafted workers	0.9225***	1.120***	1.385*	0.7134***
	(0.2762)	(0.3167)	(0.7122)	(0.2369)
No. drafted × Rank 1		-2.159***		
		(0.6828)		
No. drafted × Rank 3		8.553***		
		(3.112)		
No. drafted × Engineer			-0.6566	
			(0.6895)	
Fixed-effects				
year_num	Yes	Yes	Yes	Yes
ka_id	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
Fit statistics				
Observations	7,409	7,409	7,409	7,409

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: position × kakari × year (1938–1945). Dependent variable: remaining headcount (total workers minus drafted workers). Column (2) interacts the treatment with rank indicators (rank 1 = *yatoi*/*shokutaku*; rank 3 = *shuji*/*gishi*; rank 2 is the omitted category). Column (3) interacts with an engineer indicator. Column (4) interacts with the position's female employment share. Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 17: Lagged Hiring Test: Prior-Year Outcomes by Future Donor Status

Panel A: Unconditional Comparison

	Future Donor		Non-Donor			
	Mean	SD	Mean	SD	Diff	p-value
No. New Hires	7.447	17.243	1.867	4.813	5.580	<0.001
Avg Tenure (years)	2.724	1.294	3.237	1.908	-0.513	<0.001
Female Share	0.034	0.098	0.046	0.158	-0.012	<0.001
No. Promoted	1.509	5.618	0.375	2.169	1.134	<0.001
No. Workers	20.593	37.449	6.767	12.126	13.826	<0.001

Panel B: Conditional Comparison at $t-1$ (Position $\times$ Year FE)

	Office FE	Dept. FE
No. New Hires	4.0569***	5.1419***
	(0.4220)	(0.6665)
Avg Tenure (years)	0.0991***	0.1133***
	(0.0295)	(0.0289)
Female Share	-0.0079**	-0.0081**
	(0.0036)	(0.0032)
No. Promoted	0.9825***	1.1198***
	(0.1451)	(0.1457)
No. Workers	13.1738***	13.9780***
	(1.1855)	(1.2973)
Observations	7,134	7,235

Notes: Outcomes measured at year $t-1$; donor status at year t. Panel A reports unconditional means. Panel B reports OLS coefficients on a donor indicator with position and year fixed effects plus either office (ka) or department (kyoku) FE, controlling for log headcount at t. Standard errors clustered at the office level (1258 clusters). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 18: Characteristics of Transferred Workers

Model:	Female (1)	Tenure (2)	Rank (3)	All (4)
Variables				
Female (=1)	0.0196			0.0200
	(0.0142)			(0.0136)
Tenure (years)		0.0008		0.0012
		(0.0011)		(0.0011)
Rank (enriched)			-0.0104	-0.0112
			(0.0114)	(0.0114)
Fixed-effects				
year_num	Yes	Yes	Yes	Yes
lag_ka_id	Yes	Yes	Yes	Yes
lag_pos	Yes	Yes	Yes	Yes
Fit statistics				
Observations	49,036	49,036	49,036	49,036

Clustered (lag_office) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions (linear probability model). Dependent variable: transferred (=1 if worker moved to a different unit from prior year). Unit of observation: individual worker $\times$ year (1938–1945), restricted to workers with prior-year records. Rank is an enriched 1–5 scale serving as a productivity proxy. All specifications include year, ka (prior-year), and position (prior-year) fixed effects. Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

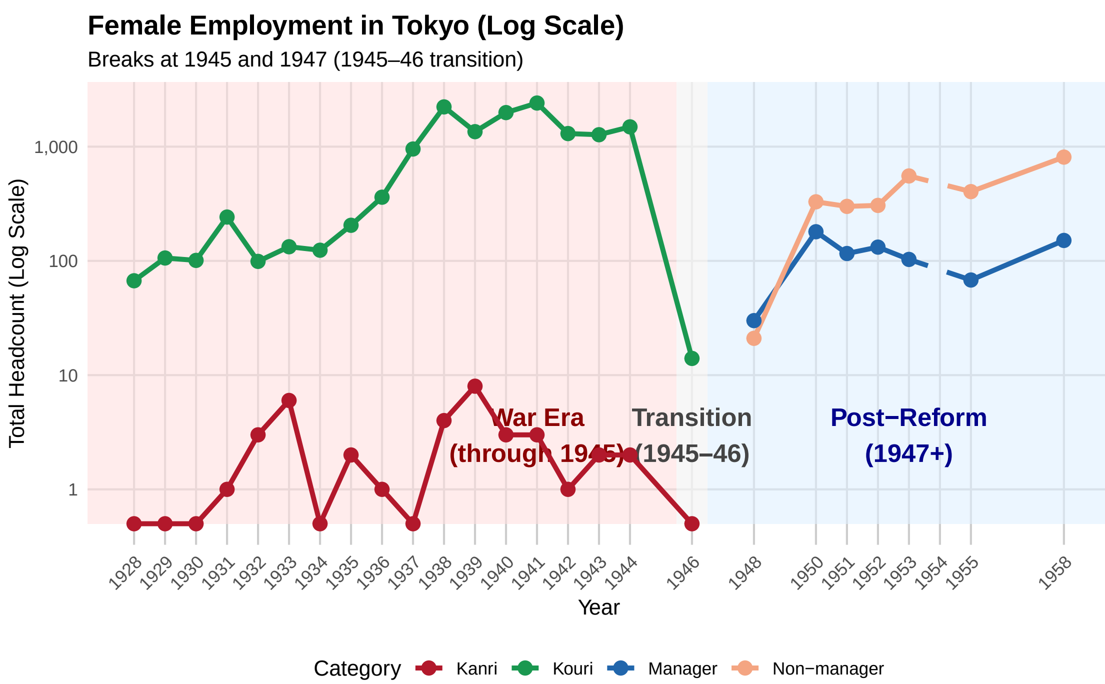
Table 19: Postwar Outcomes of Remaining Workers in Donor vs. Non-Donor Offices

	Tenure		Promotion	
Model:	(1)	(2)	(3)	(4)
Variables				
Donor office (=1)	-0.2414	-0.2117	-0.3109***	-0.3179***
	(0.2218)	(0.2308)	(0.0837)	(0.0837)
Female (=1)		-0.5761***		-0.2180*
		(0.0927)		(0.1217)
Donor × Female		-0.3260*		
		(0.1917)		
Fixed-effects				
ka_id	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
Fit statistics				
Observations	3,385	3,385	1,276	1,276
R^2	0.09043	0.09689	0.42777	0.42871

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

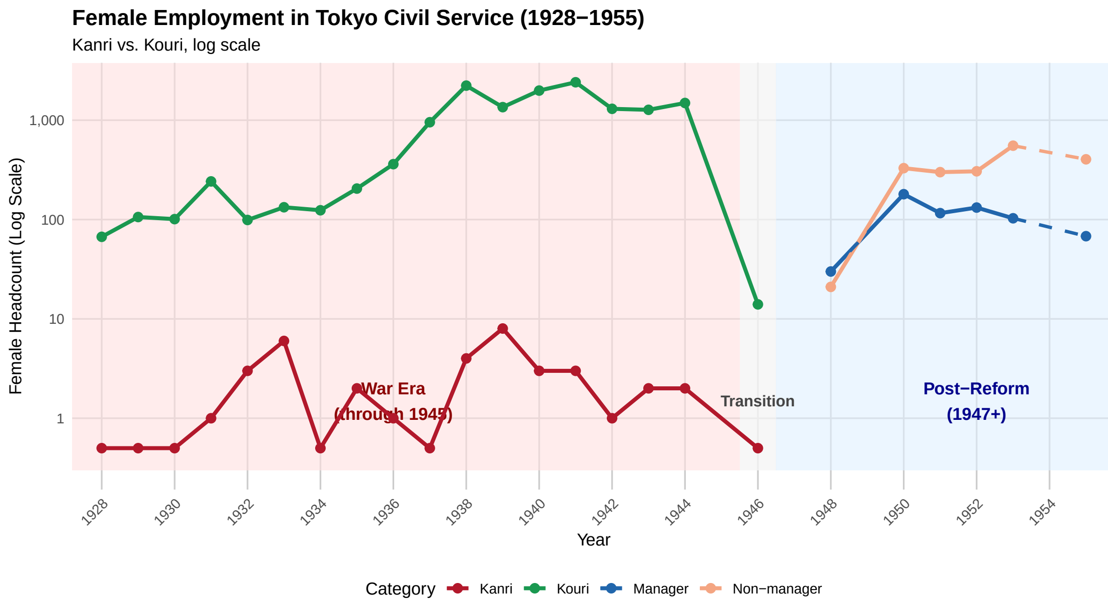
Notes: OLS regressions. Sample: non-drafted workers observed in 1944. Columns (1)–(2): dependent variable is postwar tenure (count of years appearing 1946–1955). Columns (3)–(4): dependent variable is z-scored rank change (postwar avg. minus 1944 rank). A *donor* office had at least one worker transfer out during 1938–1945. The comparison group is same-ka offices (kakari × position) that did not have transfers out. All specifications include ka and position fixed effects from 1944 and control for log cumulative male stock and kakari × position headcount (not shown). Standard errors clustered at the office level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure 3: Female Employment in Tokyo by Rank Category



Note: Headcounts are shown on a log scale; the gap reflects the 1946-1949 transition period.

Figure 4: Women in the Civil Service, 1928–1955



Note: Counts are shown on a log scale. Solid lines indicate observed data; dashed segments connect across gaps in the source records. *Kanri* denotes appointed officials; *Kouri* denotes public employees.

A.6 Women in the civil service (time series)

Table 20: Postwar Retention of 1944 Colleagues (1946–1955)

Dependent Variable:	tenure_postwar	
	Retention	
Model:	(1)	(2)
Variables		
Draft share	-0.1288	-0.0419
	(0.2345)	(0.2399)
Female (=1)		-0.5192***
		(0.1307)
Draft share × Female		-0.3388
		(0.4196)
Fixed-effects		
ka_id_1944	Yes	Yes
pos_norm_1944	Yes	Yes
Fit statistics		
Observations	3,398	3,398
R^2	0.08863	0.09526

Clustered (office_id_1944) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Sample: non-drafted workers observed in 1944. Dependent variable is count of postwar years (1946–1955) appearing in civil service. Column (2) includes gender interaction. All specifications include ka and position fixed effects from 1944 and control for log cumulative male stock. Standard errors clustered at the office level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.